Events and Probability

Lecture Notes on General Structures

Sample spaces, elementary outcomes, events, and probability models

These notes are not a solution manual. Their purpose is to build the conceptual and mathematical language needed before working on problems: how to model a random experiment, how to define elementary outcomes, how to describe events as sets, and how probabilities are assigned in finite and continuous models.

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Contents

1 What Probability Theory Tries to Model

Probability theory begins before any formula is used. The first mathematical decision is not *which probability to compute*, but *what object is being modelled.* A random situation must be translated into a precise mathematical structure.

The basic chain of ideas is:

$$\text{random experiment} \longrightarrow \text{elementary outcomes} \longrightarrow \text{sample space} \longrightarrow \text{events} \longrightarrow \text{probabilities.}$$

This order matters. If the sample space is unclear, then the events are unclear. If the events are unclear, then the probability calculation has no precise meaning.

Random experiment

A **random experiment** is a procedure whose result is not known in advance, but whose possible results can be described before the experiment is performed.

Examples of random experiments include:

- observing the status of a machine after a test,
- recording whether a message is delivered successfully,
- drawing objects from a container,
- measuring a random position or angle,
- repeating a procedure several times and recording the sequence of results.

Probability theory does not remove randomness. It gives a language for describing possible results and assigning numerical weights to events.

A common mistake

Do not start by memorizing formulas. Most mistakes in elementary probability come from an incorrectly described sample space, not from arithmetic.

2 Elementary Outcomes and Sample Spaces

2.1 Elementary outcomes

Elementary outcome

An **elementary outcome** is a complete possible result of the experiment at the chosen level of detail. It is usually denoted by ω.

The phrase *at the chosen level of detail* is important. The same real situation may be described at different levels of detail.

Example: inspecting two devices

Two devices are inspected. Each device may be classified as working (W) or faulty (F). If the order of inspection is recorded, one elementary outcome is

$$(W, F),$$

meaning that the first inspected device is working and the second inspected device is faulty. If we record only the number of faulty devices, then the possible recorded values are

$$0, 1, 2.$$

These are not the same sample spaces. The first model records a sequence; the second model records only a count.

The elementary outcome must contain enough information to decide whether any event of interest occurred. If we only record the number of faulty devices, then we cannot later ask whether the *first* device was faulty.

2.2 Sample space

Sample space

The **sample space** is the set of all elementary outcomes of the experiment. It is denoted by Ω.

For the ordered inspection of two devices,

$$\Omega = \{(W, W), (W, F), (F, W), (F, F)\}.$$

For the model that records only the number of faulty devices,

$$\Omega = \{0, 1, 2\}.$$

Both sample spaces may describe the same physical experiment, but they answer different questions.

The sample space is not unique

There is no single universal sample space for a real-world situation. The sample space depends on what is recorded and what questions we want to answer.

2.3 Choosing the correct outcome type

Before defining Ω, identify the type of elementary outcome. Common types include:

Outcome type	Typical meaning
Single symbol	One observation with finitely many categories.
Ordered tuple	Several stages are recorded, and order matters.
Set	Several objects are selected, but order is ignored.
Number	The experiment is reduced to a count or measurement.
Point in a region	A continuous outcome such as position, time, or angle.
Function or path	A whole trajectory is observed over time.

Guiding question

Ask: *What must an elementary outcome contain so that all events of interest can be recognized from it?*

3 One-Step and Multi-Step Experiments

3.1 One-step experiments

In a one-step experiment, one observation is made. If a router classifies a packet as

$$S = \{\text{accepted}, \text{delayed}, \text{rejected}\},$$

then a natural sample space is

$$\Omega = S.$$

If the three categories have probabilities $0.93, 0.05, 0.02$, then the sample space is the same but the probability assignment is not uniform.

3.2 Multi-step experiments and Cartesian products

Many experiments consist of several stages. If each stage has possible results from a set S, and the order is recorded, then the sample space is a Cartesian product:

$$\Omega = S^n = S \times S \times \cdots \times S.$$

An elementary outcome has the form

$$\omega = (s_1, s_2, \ldots, s_n),$$

where s_i is the result of stage i.

Product rule for finite staged experiments

Suppose an experiment has n stages. If stage i has m_i possible results, and every combination of stage results is possible, then the number of elementary outcomes is

$$|\Omega| = m_1 m_2 \cdots m_n.$$

In particular, if each stage has m possible results, then

$$|\Omega| = m^n.$$

Example: three server checks

A server is checked three times. At each check its status is one of

$$S = \{U, D, M\},$$

where U means up, D means down, and M means maintenance. If the order of checks is recorded, then

$$\Omega = S^3.$$

For example,

$$(U, M, D)$$

means: up at the first check, maintenance at the second check, down at the third check. The number of elementary outcomes is

$$3^3 = 27.$$

3.3 Order matters

Order matters when changing the positions of results changes the elementary outcome. For example,

$$(W, F) \neq (F, W).$$

The two outcomes describe different histories.

Order often matters when:

- results happen in time,
- positions are labelled,
- first, second, third, etc. have distinct meanings,
- a later probability may depend on an earlier result.

Order is often ignored when only the final collection matters, for example when a committee is chosen and positions are not assigned.

3.4 With replacement and without replacement

Replacement changes the set of possible later results.

With and without replacement

A selection is made **with replacement** if the selected object is returned before the next selection. The same object may then appear again.
A selection is made **without replacement** if the selected object is not returned. The same object cannot appear again in the same sequence.

Example: labelled components

Suppose a box contains four labelled components

$$C_1, C_2, C_3, C_4.$$

If two ordered selections are made with replacement, then

$$|\Omega| = 4^2 = 16.$$

The outcome (C_2, C_2) is possible.
If two ordered selections are made without replacement, then

$$|\Omega| = 4 \cdot 3 = 12.$$

The outcome (C_2, C_2) is impossible.

Same words, different model

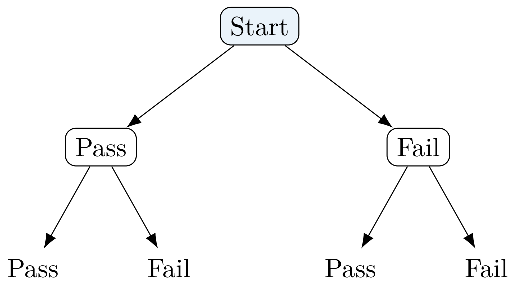
The phrase *choose two objects* is incomplete. We must know whether order is recorded and whether replacement is allowed.

4 Tree Diagrams

A tree diagram is a visual representation of a staged experiment. It does not replace the sample space; it helps construct it systematically.

Tree diagram

A **tree diagram** represents a random experiment stage by stage. Each branch corresponds to a possible result of a stage. Each complete path from the root to a leaf corresponds to one elementary outcome.



This tree describes two consecutive tests, each with two possible results. The leaves correspond to the four elementary outcomes

$$(P, P), \quad (P, F), \quad (F, P), \quad (F, F).$$

4.1 How to read a tree

1. Identify the stages of the experiment.
2. At each stage, draw all possible branches.
3. Continue until each path gives a complete result.

4. Read each root-to-leaf path as one elementary outcome.
5. Count the leaves if the sample space is finite.

A tree diagram does not automatically give probabilities

A tree shows possible outcomes. Probabilities must be assigned separately. If branches are not equally likely, counting leaves is not enough.

5 Events as Sets

5.1 Definition of an event

Event

An **event** is a subset of the sample space:

$$A \subseteq \Omega.$$

The event A occurs when the actual elementary outcome ω belongs to A.

In probability theory, statements become sets. For example, if

$$\Omega = \{(P, P), (P, F), (F, P), (F, F)\},$$

then the statement *at least one test fails* corresponds to

$$A = \{(P, F), (F, P), (F, F)\}.$$

5.2 Events, statements, and truth

An event is not a number. It is a set of elementary outcomes. A verbal statement defines an event by selecting exactly those outcomes for which the statement is true.

Example: messages and acknowledgements

A system sends two messages. Each message is either acknowledged (A) or not acknowledged (N). The sample space is

$$\Omega = \{(A, A), (A, N), (N, A), (N, N)\}.$$

The event

$$E = \text{“the first message is acknowledged”}$$

is

$$E = \{(A, A), (A, N)\}.$$

The event

$$F = \text{“both messages have the same status”}$$

is

$$F = \{(A, A), (N, N)\}.$$

5.3 Set operations and logical operations

Since events are sets, logical operations become set operations.

Logic	Set notation	Meaning
not A	A^c	outcomes for which A is false
A and B	$A \cap B$	outcomes where both events occur
A or B	$A \cup B$	outcomes where at least one event occurs
A but not B	$A \setminus B$	outcomes in A and outside B

In probability, *or* usually means inclusive or: A or B or both.

Complement, union, intersection, difference

For events $A, B \subseteq \Omega$:

$$A^c = \{\omega \in \Omega : \omega \notin A\},$$

$$A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\},$$

$$A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\},$$

$$A \setminus B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \notin B\}.$$

5.4 Disjoint events

Disjoint events

Two events A and B are **disjoint** if they cannot occur at the same time:

$$A \cap B = \varnothing.$$

Disjointness is a statement about sets, not about probabilities. If two events share no elementary outcomes, then the occurrence of one excludes the occurrence of the other.

6 Assigning Probabilities

Once Ω and the events are defined, probability is a function that assigns numbers to events.

Probability measure on a finite sample space

Let Ω be a finite sample space. A probability measure is a function

$$\mathbb{P} : 2^{\Omega} \to [0, 1]$$

such that:

1. $\mathbb{P}(\Omega) = 1$,
2. if $A \cap B = \varnothing$, then
$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

Here 2^{Ω} denotes the set of all subsets of Ω, that is, the set of all events.

6.1 Probabilities of elementary outcomes

In a finite sample space, it is enough to assign probabilities to elementary outcomes:

$$\mathbb{P}(\{\omega\}) = p_{\omega}.$$

These numbers must satisfy

$$p_\omega \geq 0, \qquad \sum_{\omega \in \Omega} p_\omega = 1.$$

Then for any event $A \subseteq \Omega$,

$$\mathbb{P}(A) = \sum_{\omega \in A} p_\omega.$$

Example: non-uniform packet status

A packet has status

$$\Omega = \{D, L, R\},$$

where D means delivered, L means delayed, and R means rejected. Suppose

$$\mathbb{P}(D) = 0.90, \qquad \mathbb{P}(L) = 0.07, \qquad \mathbb{P}(R) = 0.03.$$

The event *not delivered immediately* is

$$A = \{L, R\}.$$

Therefore

$$\mathbb{P}(A) = 0.07 + 0.03 = 0.10.$$

6.2 Equally likely outcomes

A special and very important case occurs when all elementary outcomes have the same probability.

Classical probability formula

If Ω is finite and all elementary outcomes are equally likely, then for every event $A \subseteq \Omega$,

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

This formula is correct only when elementary outcomes are equally likely. It should not be used just because the sample space is finite.

Counting is not always probability

Counting favorable outcomes and dividing by all outcomes is valid only under an equal-likelihood assumption. If outcomes have different probabilities, the event probability is obtained by summing the probabilities of its elementary outcomes.

7 Basic Probability Rules

7.1 Complement rule

Complement rule

For any event A,

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

This is often useful when the complement is simpler than the event itself.

7.2 Union rule

Inclusion-exclusion for two events

For any two events A and B,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

If A and B are disjoint, this reduces to

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

The subtraction is needed because outcomes in $A \cap B$ are counted twice in $\mathbb{P}(A) + \mathbb{P}(B)$.

7.3 Partitions

Partition of the sample space

Events $A_1, A_2, \ldots, A_k$ form a **partition** of Ω if:

1. each A_i is nonempty,
2. the events are pairwise disjoint,
3. their union is the whole sample space:

$$A_1 \cup A_2 \cup \cdots \cup A_k = \Omega.$$

If events form a partition, then

$$\mathbb{P}(A_1) + \cdots + \mathbb{P}(A_k) = 1.$$

Example: categories form a partition

Every customer account is classified as basic, standard, or premium. If each account belongs to exactly one category, then these three categories form a partition of the customer population.

8 Counting Structures Inside Probability

Counting is used in probability when the model has finitely many equally likely elementary outcomes. The goal is not to memorize many formulas, but to recognize the structure of the outcome.

8.1 Sequences

If n positions are filled independently from an alphabet of m symbols, then the number of possible sequences is

$$m^n.$$

For example, a record of n binary tests has 2^n possible sequences.

8.2 Exactly k successes in n binary positions

Suppose an outcome is a sequence of length n, and each position is either success (S) or failure (F). The number of sequences with exactly k successes is

$$\binom{n}{k}.$$

This counts which k of the n positions contain successes.

Number of binary sequences with exactly k successes

The number of sequences of length n containing exactly k successes and $n-k$ failures is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Example: alerts in monitoring windows

A monitoring system checks a service in five consecutive time windows. In each window an alert either occurs (A) or does not occur (N). The number of possible alert histories with exactly two alerts is

$$\binom{5}{2} = 10.$$

The formula counts the two positions among the five that contain alerts.

8.3 At least one

The phrase *at least one* often becomes easier through the complement:

$$\text{at least one success} = \text{not zero successes}.$$

Thus,

$$\mathbb{P}(\text{at least one success}) = 1 - \mathbb{P}(\text{no successes}).$$

Language matters

The phrases *exactly one*, *at least one*, *at most one*, and *not all* describe different events. Translate them into sets before computing.

9 Probability in Repeated Independent Trials

A common model consists of repeating the same basic experiment several times.

Independent repeated trials

Repeated trials are called **independent** if the result of one trial does not change the probabilities of the results of the other trials.

If each trial has results S and F, with

$$\mathbb{P}(S) = p, \qquad \mathbb{P}(F) = 1-p,$$

and the trials are independent, then a particular sequence with k successes and $n-k$ failures has probability

$$p^k(1-p)^{n-k}.$$

Since there are $\binom{n}{k}$ such sequences, the probability of exactly k successes is

$$\binom{n}{k}p^k(1-p)^{n-k}.$$

Binomial counting principle

In n independent binary trials with success probability p, the probability of exactly k successes is

$$\binom{n}{k}p^k(1-p)^{n-k}.$$

This formula combines two ideas: counting the positions of the successes and multiplying probabilities along independent trials.

This is not merely a formula. It expresses the structure of a repeated experiment: positions are labelled, outcomes are binary, and trials are independent.

10 Empirical Frequencies and Simulation

Probability is a mathematical model. Data from repeated experiments produce observed frequencies.

Empirical frequency

If an experiment is repeated n times and event A occurs N_A times, then the empirical relative frequency of A is

$$f_n(A) = \frac{N_A}{n}.$$

Empirical frequency is computed from observations. Probability is part of a model. These two ideas are related but not identical.

10.1 Random fluctuation

Even when the probability $\mathbb{P}(A)$ is fixed, empirical frequencies fluctuate from one sample to another. For small n, fluctuations may be large. For large n, relative frequencies often become more stable.

Law of large numbers, informal form

If an experiment is repeated independently under the same conditions, then the empirical relative frequency of an event usually becomes close to its theoretical probability when the number of repetitions is large.

The phrase *usually becomes close* is important. It does not mean that empirical frequency must equal theoretical probability exactly.

10.2 Monte Carlo simulation

A Monte Carlo simulation repeats a random experiment many times using a computer. It can be used to estimate probabilities by relative frequencies.

What simulation can and cannot do

Simulation can help build intuition and approximate probabilities. It does not replace a mathematical definition of the sample space and events. A simulation of an incorrectly modelled experiment only produces data from the wrong model.

11 Continuous Sample Spaces

Not all sample spaces are finite. Some experiments have outcomes described by real numbers or points in geometric regions.

Continuous sample space

A sample space is **continuous** when its elementary outcomes vary over an interval, region, or another uncountable set. Typical examples include positions, times, lengths, angles, and velocities.

In a continuous model, probability is not usually assigned by counting outcomes. Instead, it is assigned through length, area, volume, or density.

11.1 Uniform distribution on a region

Suppose a point is chosen uniformly at random from a region $R \subseteq \mathbb{R}^2$. Then the probability that the point lies in a subregion $A \subseteq R$ is

$$\mathbb{P}(A) = \frac{\text{area}(A)}{\text{area}(R)}.$$

Example: random point in a rectangular sensor field

A point is selected uniformly in a rectangle of width 10 and height 4. The sample space is

$$\Omega = [0, 10] \times [0, 4].$$

The event that the point lies in the left half is

$$A = [0, 5] \times [0, 4].$$

Therefore

$$\mathbb{P}(A) = \frac{5 \cdot 4}{10 \cdot 4} = \frac{1}{2}.$$

11.2 Probability of a single point

In a continuous uniform model, a single point has probability zero:

$$\mathbb{P}(\{(x_0, y_0)\}) = 0.$$

This does not mean that the point is impossible in a logical sense. It means that probability is spread continuously over infinitely many possible points.

Zero probability is not the same as impossibility

In continuous models, an event may have probability zero without being logically impossible. This is one of the major differences between finite and continuous probability spaces.

12 Geometric Probability and Needle-Type Models

Some continuous probability models are geometric. The elementary outcome is a point in a geometric region, and events are described by inequalities.

12.1 A general needle-type structure

Consider a thin segment of length L placed randomly on a plane with parallel lines separated by distance d. By symmetry, one can describe the relevant outcome using two coordinates:

- x: the distance from the center of the segment to the nearest line,
- θ: the angle between the segment and the direction of the lines.

A standard reduced sample space is

$$\Omega = \left[0, \frac{d}{2}\right] \times \left[0, \frac{\pi}{2}\right].$$

Here an elementary outcome is a pair

$$\omega = (x, \theta).$$

The event that the segment crosses a line is described by an inequality. The perpendicular distance from the center to the closer endpoint is

$$\frac{L}{2}\sin\theta.$$

Therefore crossing occurs exactly when

$$x \le \frac{L}{2}\sin\theta.$$

Buffon-type crossing probability for $L \le d$

Under the usual uniformity assumptions for x and θ, if $L \le d$, then the probability that the segment crosses a line is

$$\mathbb{P}(\text{crossing}) = \frac{2L}{\pi d}.$$

12.2 Why the formula is an area ratio

The total area of the reduced sample space is

$$\text{area}(\Omega) = \frac{d}{2} \cdot \frac{\pi}{2} = \frac{\pi d}{4}.$$

The crossing region is

$$C = \left\{ (x, \theta) : 0 \le x \le \frac{L}{2} \sin\theta, 0 \le \theta \le \frac{\pi}{2} \right\}.$$

Its area is

$$\operatorname{area}(C) = \int_0^{\pi/2} \frac{L}{2} \sin\theta \, d\theta = \frac{L}{2}.$$

Hence

$$\mathbb{P}(C) = \frac{\operatorname{area}(C)}{\operatorname{area}(\Omega)} = \frac{L/2}{\pi d/4} = \frac{2L}{\pi d}.$$

This example is important because it shows how an event can be a curved region in a continuous sample space.

What changes in continuous problems

In finite problems, probabilities may be obtained by counting elementary outcomes. In continuous geometric problems, probabilities are obtained by measuring regions. The event must still be defined as a subset of the sample space.

13 From Verbal Statements to Mathematical Events

The central skill in elementary probability is translation.

13.1 Translation pattern

For any probability question, the conceptual translation is:

$$\text{verbal statement} \longrightarrow \text{event as a subset of } \Omega \longrightarrow \text{probability of that event.}$$

This should happen even when the final calculation is short.

Minimal mathematical model

A complete probability model should specify:

1. the random experiment,
2. the elementary outcomes,
3. the sample space Ω,
4. the event or events of interest,
5. the probability assignment.

13.2 Examples of translations

Verbal phrase	Set-theoretic meaning
not A	complement A^c
at least one	complement of none
exactly one	one specified occurrence and all others absent, summed over possible positions
at most one	zero or one
all	intersection of all required conditions
A **but not** B	difference $A \setminus B$

14 Common Conceptual Errors

14.1 Confusing an elementary outcome with an event

An elementary outcome is a single complete result. An event is a set of elementary outcomes.

For example, if

$$\Omega = \{(S, S), (S, F), (F, S), (F, F)\},$$

then (S, F) is an elementary outcome, while

$$A = \{(S, F), (F, S)\}$$

is an event.

14.2 Confusing the physical experiment with the mathematical model

The physical experiment is what happens in the real world. The mathematical model is what we choose to record and analyze.

14.3 Using uniform probability without justification

A finite sample space does not imply equally likely outcomes. Equal likelihood is an assumption that must be stated or justified.

14.4 Forgetting that order may be recorded

If order is recorded, then (a, b) and (b, a) are usually different outcomes. If order is not recorded, they may represent the same outcome.

14.5 Treating a continuous sample space as finite

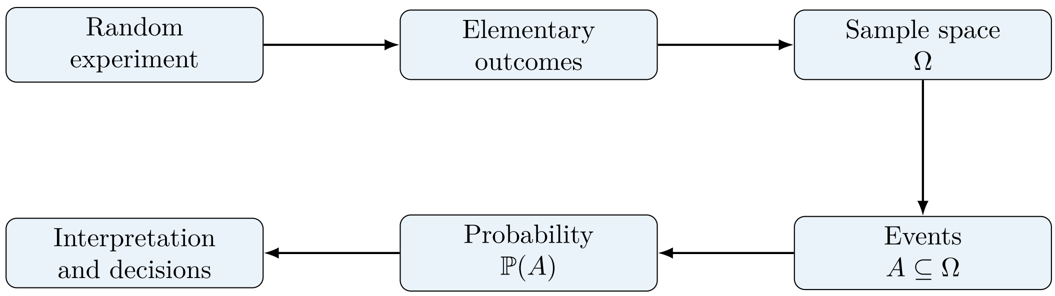
Continuous probability is not based on counting points. It is based on lengths, areas, volumes, or probability densities.

15 Summary of Core Concepts

Concept	Meaning
Random experiment	Procedure with uncertain result but describable possible outcomes.
Elementary outcome	Complete possible result at the chosen level of detail.
Sample space Ω	Set of all elementary outcomes.
Event	Subset of Ω.
Complement	Event that the original event does not occur.
Union	Event that at least one of two events occurs.
Intersection	Event that both events occur.
Disjoint events	Events with empty intersection.
Probability measure	Function assigning numbers from $[0, 1]$ to events.
Uniform finite model	Model in which all elementary outcomes are equally likely.
Empirical frequency	Observed proportion of occurrence in repeated trials.
Continuous model	Model whose outcomes form an interval, region, or other continuum.

16 A Final Conceptual Map

The main structure of the theory can be summarized as follows:



A probability calculation is meaningful only when this whole chain is clear.

What students should be able to do after these notes

After studying these notes, a student should be able to define a sample space, distinguish elementary outcomes from events, translate verbal statements into set notation, assign probabilities in finite models, recognize when counting is valid, interpret empirical frequencies, and understand why continuous models require geometric or density-based probability rather than counting.